Color Recognition

This part is mainly to prepare for future functional features. Main steps:

- Convert the RGB image to be detected into HSV image
- Define an object of Mat type: mask
- Define color upper and lower bounds

Where the upper bound is a Scalar object containing three values: hmin, smin, vmin, representing the minimum values of the three HSV elements;

And the lower bound is also a Scalar object containing three values: hmax, smax, vmax, representing the maximum values of the three HSV elements.

- Use the inRange function to detect whether each pixel of the src image is between lowerb and upperb

If so, this pixel is set to 255 and saved in the mask image, otherwise it is 0.

1. Basic Principles

Commonly used models in digital image processing are RGB (Red, Green, Blue) model and HSV (Hue, Saturation, Value). RGB is widely used in color monitors and color video cameras, and our usual images are generally RGB models. The HSV model is more in line with the way humans describe and explain colors. The color description of HSV is natural and very intuitive to humans. Another reason for choosing the HSV model is that RGB channels cannot well reflect the specific color information of objects, while compared to RGB space, HSV space can very intuitively express the brightness, hue, and vividness of colors, making it convenient to compare colors.

2. HSV Model

HSV (Hue, Saturation, Value) is a color space created by A. R. Smith in 1978 based on the intuitive characteristics of colors, also known as the Hexcone Model. The color parameters in this model are: Hue (H), Saturation (S), and Value (V).

H: 0 — 180

S: 0 — 255

V: 0 — 255

- HSV Parameter Table:

	Black	Gray	White	Red	Orange	Yellow	Green	Cyan	Blue	Purple	
H_min	0	0	0	0	156	11	26	35	78	100	125
H_max	180	180	180	10	180	25	34	77	99	124	155
S_min	0	0	0	43	43	43	43	43	43	43	
S_max	255	43	30	255	255	255	255	255	255	255	
V_min	0	0	0	46	46	46	46	46	46	46	
V_max	46	220	255	255	255	255	255	255	255	255	

3. Main Code

- Start Docker

After entering the main board desktop, open the terminal and run the following command to start the corresponding Dofbot container:

```
./Docker_Ros.sh
```

Then enter the following command in the docker terminal to start jupyter-lab,

```
cd
jupyter-lab --allow-root
```

Then enter the main board's IP address plus :9999 in the browser,

```
IP:9999         // Example: 192.168.1.11:9999
```

Code path: /root/Dofbot/6.AI_Visuall/5.Color recognition.ipynb

The following code content needs to be executed step by step according to each step, and cannot be run all at once. Running the last unit will directly exit the thread.

```
#bgr8 to jpeg format
import enum
import cv2
def bgr8_to_jpeg(value, quality=75):
    return bytes(cv2.imencode('.jpg', value)[1])
```

```
# Camera component display
import traitlets
import ipywidgets.widgets as widgets
import time
# Thread function operation library
import threading
import inspect
import ctypes

origin_widget = widgets.Image(format='jpeg', width=320, height=240)
mask_widget = widgets.Image(format='jpeg',width=320, height=240)
result_widget = widgets.Image(format='jpeg',width=320, height=240)

# Create a horizontal box container to place image widgets side by side
image_container = widgets.HBox([origin_widget, mask_widget, result_widget])
# image_container = widgets.Image(format='jpeg', width=600, height=500)
display(image_container)
```

Get HSV values of colors

```
def get_color(img):
    H = []
    color_name={}
    img = cv2.resize(img, (640, 480), )
    # Convert color image to HSV
```

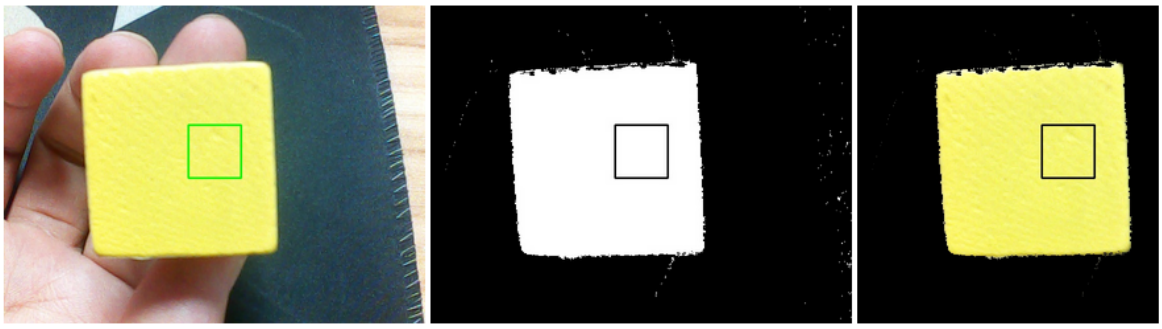
```
    HSV = cv2.cvtColor(img, cv2.COLOR_BGR2HSV)
    # Draw rectangle
    cv2.rectangle(img, (280, 180), (360, 260), (0, 255, 0), 2)
    # Take out H, S, V values of each row and column in sequence and put them in
the container
    for i in range(280, 360):
        for j in range(180, 260): H.append(HSV[j, i][0])
    # Calculate H, S, V maximum and minimum respectively
    H_min = min(H);H_max = max(H)
#     print(H_min,H_max)
    # Judge color
    if H_min >= 0 and H_max <= 10 or H_min >= 156 and H_max <= 180:
color_name['name'] = 'red'
    elif H_min >= 26 and H_max <= 34: color_name['name'] = 'yellow'
    elif H_min >= 35 and H_max <= 78: color_name['name'] = 'green'
    elif H_min >= 100 and H_max <= 124: color_name['name'] = 'blue'
    return img, color_name
```

Main process: Recognize red, green, blue, yellow colors.

```
import cv2
import numpy as np
import ipywidgets.widgets as widgets


cap = cv2.VideoCapture(0)
cap.set(3, 640)
cap.set(4, 480)
cap.set(5, 30)  # Set frame rate
cap.set(cv2.CAP_PROP_FOURCC, cv2.VideoWriter.fourcc('M', 'J', 'P', 'G'))

# Default select red, the program will automatically switch colors based on the
color detected in the box
# Red range
color_lower = np.array([0, 43, 46])
color_upper = np.array([10, 255, 255])


def Color_Recongnize():

    while(1):
        # get a frame and show Get video frame and convert to HSV format, use
cvtColor() to convert BGR format to HSV format, parameter is cv2.COLOR_BGR2HSV.
        ret, frame = cap.read()
        frame, color_name = get_color(frame)
        if len(color_name)==1:
            global color_lower
            global color_upper

            if color_name['name'] == 'yellow':
                color_lower = np.array([26, 43, 46])
                color_upper = np.array([34, 255, 255])

            elif color_name['name'] == 'red':
                color_lower = np.array([0, 43, 46])
                color_upper = np.array([10, 255, 255])
```

```
            elif  color_name['name'] == 'green':
                color_lower = np.array([35, 43, 46])
                color_upper = np.array([77, 255, 255])

            elif color_name['name'] == 'blue':
                color_lower=np.array([100, 43, 46])
                color_upper = np.array([124, 255, 255])

        origin_widget.value = bgr8_to_jpeg(frame)
        #cv2.imshow('Capture', frame)

        # change to hsv model
        hsv = cv2.cvtColor(frame, cv2.COLOR_BGR2HSV)

        # get mask Use the inRange() function and the upper and lower bounds of
the blue range in the HSV model to get the mask. The blue part in the mask in the
original video will be turned white, and other parts black.
        mask = cv2.inRange(hsv, color_lower, color_upper)
        #cv2.imshow('Mask', mask)
        mask_widget.value = bgr8_to_jpeg(mask)


        # detect blue Perform bitwise AND operation between mask and original
video frame, which will replace the white in the mask with the real image:
        res = cv2.bitwise_and(frame, frame, mask=mask)
        #cv2.imshow('Result', res)
        result_widget.value = bgr8_to_jpeg(res)

        time.sleep(0.01)

    cap.release()
    #cv2.destroyAllWindows()
```

After the program block runs, see the camera component display,



4. Debugging Method

If a certain color cannot be recognized during the color recognition process, you can modify the color judgment values according to the following steps:

Uncomment `print(H_min,H_max)`, then put the unrecognized block into the recognition box, look at the H_min, H_max values output by the program, and then replace the color judgment values with these values.

```
def get_color(img):
    H = []
    color_name={}
```

```
    img = cv2.resize(img, (640, 480), )

    HSV = cv2.cvtColor(img, cv2.COLOR_BGR2HSV)

    cv2.rectangle(img, (280, 180), (360, 260), (0, 255, 0), 2)

    for i in range(280, 360):
        for j in range(180, 260): H.append(HSV[j, i][0])

    H_min = min(H);H_max = max(H)
    # print(H_min,H_max)

    # Color judgment
    if H_min >= 0 and H_max <= 10 or H_min >= 156 and H_max <= 180:
color_name['name'] = 'red'
    elif H_min >= 25 and H_max <= 30: color_name['name'] = 'yellow'
    elif H_min >= 35 and H_max <= 78: color_name['name'] = 'green'
    elif H_min >= 100 and H_max <= 124: color_name['name'] = 'blue'
    return img, color_name
```